Formal Frame, Syntactic Support, and Real Determination in Finite Boolean Functions

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Abstract

We study a finite typed Boolean function

$$f : \mathbf{2}^N \longrightarrow \mathbf{2}$$

together with its determinational support R_f and, for a chosen term presentation τ, its syntactic support S_τ. Every such presentation satisfies

$$R_f \subseteq S_\tau \subseteq N,$$

which separates argument-frame membership, literal syntactic presence, and determinational power.

For the fixed function f, let $\mathfrak{S}_N(f)$ denote the family of syntactic supports of all Boolean terms representing f on the frame N. We prove that

$$\mathfrak{S}_N(f) = [R_f, N]_\subseteq,$$

so every set of coordinates between the determinational support and the formal frame occurs as the exact syntactic support of some presentation. Equivalently, the presentation-support family is order-isomorphic to $\mathcal{P}(N \setminus R_f)$.

Variant 11,

$$f_{11}(p, q) = \neg q,$$

provides a minimal-arity Boolean model containing one determinationally real coordinate and one determinationally fictitious coordinate. Its alternative term presentations exhibit the two endpoint supports $R_{f_{11}}$ and N.

The determinational core

$$f_{R_f} : \mathbf{2}^{R_f} \longrightarrow \mathbf{2}$$

is uniquely determined by

$$f = \mathrm{Cyl}^N_{R_f}(f_{R_f}).$$

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It contains all information required to determine the values of f, but it does not by itself determine the ambient frame N or a chosen term presentation τ. Thus determinational completeness, formal-frame membership, and syntactic visibility remain distinct aspects of the same typed function.

Keywords: Boolean functions, essential variables, formal frame, syntactic support, real determination, term representation.
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1 Introduction

A Boolean function may be formally given on a frame that is wider than the set of coordinates on which its value actually depends. This creates a distinction between what belongs to the argument structure of a function and what has determinational power within that structure.

A further distinction appears when the same typed function is represented by a Boolean term. A coordinate may belong to the formal frame without occurring in a particular term, or it may occur literally in the term without having any power to change the value of the function. Frame membership, syntactic presence, and determinational power therefore answer different questions.

The real/fictitious distinction used here develops the distinction introduced in Paper III of *Minimal Reality, Volume I: The Minimal Structure of Change* [4]. Formally, it coincides with the classical essential/inessential-variable distinction for Boolean functions [1, 3]. The theory of essential variables and essential arity has subsequently been developed in a broader setting; see, for example, [2]. The present construction keeps the ambient typed frame fixed and studies, within that frame, the full family of syntactic supports of equivalent term presentations.

For a finite typed Boolean function

$$f : \mathbf{2}^N \longrightarrow \mathbf{2},$$

we distinguish the formal frame N, the determinational support R_f, and, for a chosen term presentation τ, the syntactic support S_τ. Every representation satisfies

$$R_f \subseteq S_\tau \subseteq N.$$

The significance of this relation lies in the separation it records: what belongs to the formal argument frame, what is made literally present by a presentation, and what can affect the output need not coincide.

The inclusion chain has a stronger global consequence. If $\mathfrak{S}_N(f)$ denotes the family of syntactic supports of all terms representing the fixed typed function f, then

$$\mathfrak{S}_N(f) = [R_f, N]_\subseteq.$$

Thus the determinational support is the lower boundary of all admissible presentation supports, the formal frame is the upper boundary, and every intermediate support is realizable.

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Equivalently, the presentation-dependent part of the syntactic support is exactly an arbitrary subset of $N \setminus R_f$.

Variant 11,

$$f_{11}(p, q) = \neg q,$$

provides a minimal-arity Boolean model containing one determinationally real coordinate and one determinationally fictitious coordinate. Different presentations of this fixed typed function show that the fictitious coordinate may be excluded from the term or made literally present in it while remaining outside the determinational support.

The same distinction persists at the level of the determinational core. The factorization

$$f = \mathrm{Cyl}^N_{R_f}(f_{R_f})$$

shows that the values of f are determined entirely by the coordinates in R_f. At the same time, the core alone does not determine the ambient frame N or a chosen presentation of the function. Data sufficient to determine every value therefore need not specify the full formal situation in which the typed function is given.

The analysis developed here is entirely static. It concerns the relations among frame, presentation, and determination for a fixed typed function. Its purpose is to isolate the formal structure of that separation without introducing any dynamical or additional determinational mechanism.

2 Frame and Determination

Let N be a finite labelled set of coordinates. Write

$$\mathbf{2} = \{0, 1\}$$

and

$$\mathbf{2}^N = \{x : x : N \longrightarrow \mathbf{2}\}.$$

Let

$$f : \mathbf{2}^N \longrightarrow \mathbf{2}.$$

The function is defined on the frame N, but its value need not depend on every coordinate belonging to that frame.

A coordinate $i \in N$ is *determinationally real* for f if there exist $x, y \in \mathbf{2}^N$ such that

$$x\big|_{N\setminus\{i\}} = y\big|_{N\setminus\{i\}}$$

and

$$f(x) \neq f(y).$$

The determinational support of f is

$$R_f = \left\{ i \in N : \exists x, y \in \mathbf{2}^N \left(x\big|_{N\setminus\{i\}} = y\big|_{N\setminus\{i\}} \ \wedge \ f(x) \neq f(y) \right) \right\}.$$

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Formally, R_f is the classical essential support of f [3]. In the present terminology, it records exactly those coordinates whose variation, with all other coordinates fixed, can change the output.

A coordinate $i \in N \setminus R_f$ is called *determinationally fictitious* for f.

By construction,

$$R_f \subseteq N.$$

The inclusion may be strict:

$$R_f \subsetneq N.$$

In that case, some coordinates belong to the formal frame of the function while having no power to alter its value. They are present in the type of f, but they do not participate in its determination.

Thus N and R_f record two different aspects of the same typed function. The frame records which coordinates are formally available as arguments; the determinational support records which coordinates can make a difference to the output.

This gives the first separation:

the frame may be wider than determination.

3 Term and Syntactic Support

Let $\mathcal{L}$ be the standard Boolean term language over the variable symbols indexed by N, generated using the constants $0, 1$ and the connectives

$$\neg, \qquad \wedge, \qquad \vee,$$

with their usual two-valued truth-functional semantics. This language is functionally complete; hence every Boolean function

$$f : \mathbf{2}^N \longrightarrow \mathbf{2}$$

admits an $\mathcal{L}$-term representation.

For every $\mathcal{L}$-term τ, its interpretation on the frame N is a Boolean function

$$[\![\tau]\!]^{\mathcal{L}}_N : \mathbf{2}^N \longrightarrow \mathbf{2}.$$

Let τ be an $\mathcal{L}$-term representing the fixed typed function

$$f : \mathbf{2}^N \longrightarrow \mathbf{2}.$$

Thus

$$[\![\tau]\!]^{\mathcal{L}}_N = f.$$

The term τ is a particular presentation of f within the frame N. Its choice does not alter the determinational rule already fixed by the function.

The *syntactic support* of τ is

$$S_\tau = \{i \in N : i \text{ occurs literally in } \tau\}.$$

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Unlike the determinational support R_f, the set S_τ depends on the chosen term. It records which coordinates are made literally present by that presentation. It does not record the full syntactic form of τ, its length, its term tree, or the history by which the term was obtained.

Since every variable symbol occurring in τ belongs to the frame,

$$S_\tau \subseteq N.$$

Proposition 3.1 (Term-support invariance)**.** *Let τ be an $\mathcal{L}$-term and let $x, y \in \mathbf{2}^N$. If*

$$x\big|_{S_\tau} = y\big|_{S_\tau},$$

then

$$[\![\tau]\!]^{\mathcal{L}}_N(x) = [\![\tau]\!]^{\mathcal{L}}_N(y).$$

Proof. The statement follows by structural induction on τ. A constant term is independent of every coordinate. If τ is the variable symbol i, then

$$S_\tau = \{i\},$$

and equality of the restrictions to S_τ gives $x(i) = y(i)$. For compound terms, the induction hypothesis applies to each immediate subterm, and the truth-functional operations $\neg, \wedge, \vee$ preserve equality of the corresponding values. Therefore the value of τ depends only on the coordinates in S_τ. □

Proposition 3.2 (Determinational support is contained in syntactic support)**.** *If*

$$[\![\tau]\!]^{\mathcal{L}}_N = f,$$

then

$$R_f \subseteq S_\tau.$$

Proof. Suppose that $i \notin S_\tau$. Let $x, y \in \mathbf{2}^N$ satisfy

$$x\big|_{N\setminus\{i\}} = y\big|_{N\setminus\{i\}}.$$

Since $i \notin S_\tau$, the points x and y agree on every coordinate in S_τ. By term-support invariance,

$$[\![\tau]\!]^{\mathcal{L}}_N(x) = [\![\tau]\!]^{\mathcal{L}}_N(y).$$

Because

$$[\![\tau]\!]^{\mathcal{L}}_N = f,$$

we obtain

$$f(x) = f(y).$$

Hence $i \notin R_f$. Therefore

$$R_f \subseteq S_\tau.$$

□

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This inclusion is the contrapositive of the standard representation-theoretic observation that a variable absent from a Boolean expression cannot affect the function represented by that expression; see Crama and Hammer [3, Sec. 1.8].

Every term representation of f thus satisfies

$$\boxed{R_f \subseteq S_\tau \subseteq N.}$$

The three sets record distinct aspects of the same fixed typed function and its chosen presentation: N gives the formal frame of f, R_f its determinational support, and S_τ the coordinates made literally present by τ.

The presentation does not change determination; S_τ is a controlled trace of presentation choice.

4 Central Separation

The relation

$$R_f \subseteq S_\tau \subseteq N$$

brings together three distinct structural aspects of the same fixed typed function and its chosen presentation.

The frame N records which coordinates belong to the formal argument structure of f. The syntactic support S_τ records which of those coordinates occur literally in the chosen term. The determinational support R_f records which coordinates can change the output.

These are different questions:

$$\begin{aligned} N &: \text{what belongs to the formal frame?} \\ S_\tau &: \text{what is literally present in the chosen term?} \\ R_f &: \text{what can change the value of the function?} \end{aligned}$$

Their answers need not coincide.

A coordinate may belong to the frame without belonging to the determinational support. It may also occur literally in the chosen term while remaining outside R_f. Thus a coordinate may be present in the formal frame, or even explicitly present in the term, without having any power to alter the value of the function.

What belongs to the frame, what is shown by the chosen presentation, and what acts on the output are therefore distinct aspects of the same formal situation.

This is the central separation:

argument presence, syntactic presence, and determinational power are not the same.

In particular, lack of determinational power does not imply lack of structural presence.

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5 Variant 11

The separation established above becomes concrete in Variant 11.

Let

$$N = \{p, q\},$$

and consider the typed Boolean function

$$f_{11} : \mathbf{2}^N \longrightarrow \mathbf{2}$$

defined by

$$f_{11}(p, q) = \neg q.$$

Changing q can change the value of the function, whereas changing p cannot. Hence

$$R_{f_{11}} = \{q\},$$

so that

$$R_{f_{11}} \subsetneq N.$$

Consider first the term

$$\tau_0 = \neg q.$$

It represents f_{11} in the fixed frame N:

$$[\![\tau_0]\!]^{\mathcal{L}}_N = f_{11}.$$

Its syntactic support is

$$S_{\tau_0} = \{q\} = R_{f_{11}}.$$

Thus p belongs to the frame but is not made literally present by this presentation:

$$p \in N, \qquad p \notin S_{\tau_0}, \qquad p \notin R_{f_{11}}.$$

Now consider the term

$$\tau_1 = (p \vee \neg p) \wedge \neg q.$$

Since

$$p \vee \neg p = 1,$$

it represents the same typed function in the same frame:

$$[\![\tau_1]\!]^{\mathcal{L}}_N = f_{11}.$$

Its syntactic support, however, is

$$S_{\tau_1} = \{p, q\} = N.$$

The determinational support remains

$$R_{f_{11}} = \{q\}.$$

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Therefore

$$p \in N, \qquad p \in S_{\tau_1}, \qquad p \notin R_{f_{11}}.$$

The coordinate p remains present in the formal frame and is now also literally present in the chosen term, while still having no power to alter the value of the function.

$$\boxed{p \text{ is present, but does not act on the value.}}$$

The two terms

$$\tau_0 = \neg q \qquad \text{and} \qquad \tau_1 = (p \vee \neg p) \wedge \neg q$$

represent the same typed function in the same frame:

$$[\![\tau_0]\!]^{\mathcal{L}}_N = [\![\tau_1]\!]^{\mathcal{L}}_N = f_{11},$$

while their syntactic supports differ:

$$S_{\tau_0} = R_{f_{11}} \subsetneq N,$$

whereas

$$R_{f_{11}} \subsetneq S_{\tau_1} = N.$$

The determinational rule has not changed. What has changed is the presentation, and with it the extent to which the separation between frame and determination is made visible in the term.

Variant 11 thus exhibits the central distinction in a minimal-arity Boolean setting containing both a determinationally real coordinate and a determinationally fictitious coordinate: formal and syntactic presence may exceed determinational power without changing the typed function.

Remark 5.1 (Minimality of Variant 11 in arity). Any typed Boolean function with at least one determinationally real coordinate $r \in R_f$ and at least one determinationally fictitious coordinate $s \in N \setminus R_f$ must satisfy $r \neq s$, and therefore

$$|N| \geq 2.$$

Variant 11 attains this lower bound, since

$$N = \{p, q\}, \qquad R_{f_{11}} = \{q\}, \qquad p \in N \setminus R_{f_{11}}.$$

6 Presentation-Support Interval and Visibility Grammar

Throughout this section, $\mathcal{L}$ denotes the fixed Boolean term language introduced above. The dependence of the presentation-support family on $\mathcal{L}$ is suppressed in the notation.

For a fixed typed function

$$f : \mathbf{2}^N \longrightarrow \mathbf{2},$$

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define the set of its term presentations on the frame N by

$$\mathrm{Rep}_N(f) = \left\{\tau : [\![\tau]\!]_N^{\mathcal{L}} = f\right\}.$$

Define the corresponding family of syntactic supports by

$$\mathfrak{S}_N(f) = \{S_\tau : \tau \in \mathrm{Rep}_N(f)\}.$$

The inclusion

$$R_f \subseteq S_\tau \subseteq N$$

holds for every $\tau \in \mathrm{Rep}_N(f)$. The converse question is whether every set between R_f and N can occur as the exact syntactic support of some presentation of the same typed function.

Theorem 6.1 (Presentation-support interval)**.** *For every finite typed Boolean function*

$$f : \mathbf{2}^N \longrightarrow \mathbf{2},$$

one has

$$\boxed{\mathfrak{S}_N(f) = [R_f, N]_\subseteq = \{S \subseteq N : R_f \subseteq S\}.}$$

Proof. If $\tau \in \mathrm{Rep}_N(f)$, then the support inclusion already proved gives

$$R_f \subseteq S_\tau \subseteq N.$$

Hence

$$\mathfrak{S}_N(f) \subseteq [R_f, N]_\subseteq.$$

For the converse inclusion, first construct a presentation whose syntactic support is exactly R_f. Choose an $\mathcal{L}$-term σ representing f on N.

Let $i \in S_\sigma \setminus R_f$. Replace every occurrence of the variable symbol i in σ by the constant 0, obtaining a term $\sigma[i := 0]$. Because $i \notin R_f$, changing only coordinate i does not change the value of f. Therefore

$$[\![\sigma[i := 0]]\!]_N^{\mathcal{L}} = f.$$

Repeating this elimination for every coordinate in $S_\sigma \setminus R_f$ yields a term ρ representing f such that

$$S_\rho \subseteq R_f.$$

But every term representing f satisfies

$$R_f \subseteq S_\rho.$$

Consequently,

$$S_\rho = R_f.$$

This argument also covers $R_f = \varnothing$; in that case f is constant and the elimination process yields a variable-free term representing it.

Now let

$$R_f \subseteq S \subseteq N.$$

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Define

$$\tau_S = \rho \wedge \bigwedge_{i \in S \setminus R_f} (i \vee \neg i),$$

with the empty conjunction interpreted as the constant 1. Since each factor $i \vee \neg i$ is identically equal to 1,

$$[\![\tau_S]\!]_N^{\mathcal{L}} = f.$$

Moreover,

$$S_{\tau_S} = S_\rho \cup (S \setminus R_f) = R_f \cup (S \setminus R_f) = S.$$

Thus every $S \in [R_f, N]_\subseteq$ belongs to $\mathfrak{S}_N(f)$, and therefore

$$\mathfrak{S}_N(f) = [R_f, N]_\subseteq.$$

□

Corollary 6.2 (Boundary reconstruction)**.** *For every finite typed Boolean function $f : \mathbf{2}^N \to \mathbf{2}$,*

$$\boxed{R_f = \bigcap_{\tau \in \mathrm{Rep}_N(f)} S_\tau}$$

and

$$\boxed{N = \bigcup_{\tau \in \mathrm{Rep}_N(f)} S_\tau.}$$

Proof. The interval theorem shows that R_f and N themselves occur as members of $\mathfrak{S}_N(f)$, and every member of the family lies between them. □

Corollary 6.3 (Presentation-support Boolean lattice)**.** *The map*

$$\Phi : \mathfrak{S}_N(f) \longrightarrow \mathcal{P}(N \setminus R_f), \qquad \Phi(S) = S \setminus R_f,$$

is an order isomorphism with inverse

$$E \longmapsto R_f \cup E.$$

Consequently,

$$\boxed{\mathfrak{S}_N(f) \cong \mathcal{P}(N \setminus R_f)}$$

as partially ordered sets under inclusion, and

$$\boxed{|\mathfrak{S}_N(f)| = 2^{|N \setminus R_f|}.}$$

In particular, $\mathfrak{S}_N(f)$, ordered by inclusion, is a finite Boolean lattice with bottom element R_f, top element N, and rank

$$|N \setminus R_f|.$$

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Equivalently, every presentation support admits the unique decomposition

$$S_\tau = R_f \cup E_\tau, \qquad E_\tau \subseteq N \setminus R_f,$$

and every subset of $N \setminus R_f$ occurs as such an E_τ. The fixed part R_f is forced by determination; the presentation-dependent part lies entirely inside the excess of the frame over the determinational support.

This interval structure gives a precise form to the visibility grammar. If

$$R_f = N,$$

then

$$\mathfrak{S}_N(f) = \{N\},$$

so every presentation has

$$S_\tau = R_f = N.$$

Suppose now that

$$R_f \subsetneq N.$$

The two endpoint configurations are always realizable:

$$R_f = S_\tau \subsetneq N$$

and

$$R_f \subsetneq S_\tau = N.$$

A strict intermediate configuration

$$R_f \subsetneq S_\tau \subsetneq N$$

is realizable exactly when $N \setminus R_f$ has at least two elements. In that case, the strict intermediate supports correspond precisely to the nonempty proper subsets of $N \setminus R_f$.

Thus the qualitative forms

$$R_f = S_\tau \subsetneq N, \qquad R_f \subsetneq S_\tau \subsetneq N, \qquad R_f \subsetneq S_\tau = N$$

are not merely descriptive possibilities. They are positions inside the full presentation-support interval $[R_f, N]_\subseteq$, whose members are all realized by terms representing the same typed function.

7 Core and Structural Boundary

Let

$$f : \mathbf{2}^N \longrightarrow \mathbf{2}$$

be a fixed typed Boolean function with determinational support R_f.

For every $M \subseteq N$, let

$$\pi^N_M : \mathbf{2}^N \longrightarrow \mathbf{2}^M$$

denote the canonical restriction map

$$\pi^N_M(x) = x\big|_M.$$

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Proposition 7.1 (Determinational core factorization)**.** *There exists a unique Boolean function*

$$f_{R_f} : \mathbf{2}^{R_f} \longrightarrow \mathbf{2}$$

such that

$$f = f_{R_f} \circ \pi^N_{R_f}.$$

Proof. First we show that f is constant on every fiber of $\pi^N_{R_f}$.

Let $x, y \in \mathbf{2}^N$ satisfy

$$x\Big|_{R_f} = y\Big|_{R_f}.$$

Set

$$D = \{i \in N : x(i) \neq y(i)\}.$$

Then

$$D \subseteq N \setminus R_f.$$

Since N is finite, write

$$D = \{i_1, \ldots, i_k\}.$$

Starting with $z_0 = x$, construct z_j from z_{j-1} by replacing only the coordinate i_j with the value $y(i_j)$. Then

$$z_k = y,$$

and for every j,

$$z_{j-1}\Big|_{N\setminus\{i_j\}} = z_j\Big|_{N\setminus\{i_j\}}.$$

Because $i_j \notin R_f$, changing only coordinate i_j cannot change the value of f. Hence

$$f(z_{j-1}) = f(z_j)$$

for every j, and therefore

$$f(x) = f(y).$$

Thus f is constant on every fiber of $\pi^N_{R_f}$.

For $u \in \mathbf{2}^{R_f}$, choose any $x \in \mathbf{2}^N$ satisfying

$$x\Big|_{R_f} = u$$

and define

$$f_{R_f}(u) = f(x).$$

The preceding argument shows that this definition is independent of the chosen extension x. Hence f_{R_f} is well defined and

$$f = f_{R_f} \circ \pi^N_{R_f}.$$

For uniqueness, suppose

$$g : \mathbf{2}^{R_f} \longrightarrow \mathbf{2}$$

also satisfies

$$f = g \circ \pi^N_{R_f}.$$

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For every $u \in \mathbf{2}^{R_f}$, choose an extension $x \in \mathbf{2}^N$ with

$$x\Big|_{R_f} = u.$$

Then

$$g(u) = f(x) = f_{R_f}(u).$$

Thus

$$g = f_{R_f}.$$

□

The function

$$f_{R_f} : \mathbf{2}^{R_f} \longrightarrow \mathbf{2}$$

is called the *determinational core* of f.

For a Boolean function

$$g : \mathbf{2}^M \longrightarrow \mathbf{2}, \qquad M \subseteq N,$$

define its cylindrical extension to the frame N by

$$\mathrm{Cyl}_M^N(g) = g \circ \pi_M^N.$$

The factorization through the determinational support can therefore be written as

$$\boxed{f = \mathrm{Cyl}_{R_f}^N(f_{R_f}).}$$

The core determines every value of f once the restriction to R_f is given. The ambient frame N, however, remains part of the type of the function.

If

$$R_f \subsetneq N,$$

then

$$f_{R_f} : \mathbf{2}^{R_f} \longrightarrow \mathbf{2}$$

and

$$f : \mathbf{2}^N \longrightarrow \mathbf{2}$$

are not the same typed function. Their domains are different, even though the value of f depends only on the coordinates in R_f.

The core does not by itself determine the ambient frame N. In particular, it does not determine the excess

$$N \setminus R_f.$$

Nor does the core determine a chosen term presentation τ of f, or its syntactic support S_τ.

In this precise sense, the core is determinationally complete: it contains all data required for the determination of values. It does not, however, encode the full frame-and-presentation data.

A precise boundary is therefore already present within the static formal picture: determinational completeness need not amount to a complete specification of the formal situation in which the typed function is given.

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8 Conclusion

The relation

$$R_f \subseteq S_\tau \subseteq N$$

separates three questions that need not have the same answer. The frame N records what belongs to the formal argument structure of the function. The syntactic support S_τ records what is literally present in a chosen term. The determinational support R_f records what can change the value of the function.

The separation is stronger than the inclusion chain alone suggests. For the fixed typed function f, the family of syntactic supports of all term presentations is exactly

$$\boxed{\mathfrak{S}_N(f) = [R_f, N]_\subseteq.}$$

Hence every presentation support has the form

$$S_\tau = R_f \cup E_\tau, \qquad E_\tau \subseteq N \setminus R_f,$$

and every such subset E_τ occurs. Equivalently,

$$\mathfrak{S}_N(f) \cong \mathcal{P}(N \setminus R_f).$$

The determinational support is therefore the necessary part of every presentation support, while the excess $N \setminus R_f$ is the entire domain of syntactic-support variation.

Variant 11 exhibits this distinction at minimal arity. For

$$f_{11}(p, q) = \neg q,$$

one has

$$R_{f_{11}} = \{q\} \qquad \text{and} \qquad N \setminus R_{f_{11}} = \{p\}.$$

The presentations

$$\tau_0 = \neg q \qquad \text{and} \qquad \tau_1 = (p \vee \neg p) \wedge \neg q$$

realize the two endpoint supports

$$S_{\tau_0} = R_{f_{11}} \qquad \text{and} \qquad S_{\tau_1} = N.$$

Thus p may be absent from a presentation or literally present in it while remaining determinationally fictitious.

The same separation appears at the level of the determinational core. For

$$f : \mathbf{2}^N \longrightarrow \mathbf{2},$$

the unique factorization

$$f = \mathrm{Cyl}^N_{R_f}(f_{R_f})$$

shows that every value of f is determined by

$$f_{R_f} : \mathbf{2}^{R_f} \longrightarrow \mathbf{2}.$$

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At the same time, the ambient frame N remains part of the type of f, and the core alone does not determine either that frame or a chosen term presentation of the function.

In this precise sense, determinational completeness does not exhaust the formal situation. The function fixes the lower boundary R_f, the type fixes the upper boundary N, and presentation varies throughout the entire interval between them without introducing new determinational power.

What belongs to the formal frame, what is syntactically shown, and what is determinationally active need not coincide; the admissible syntactic supports fill exactly the interval between determination and frame.

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